Linear A: Anti-Circularity Experiment Suite

Seven experiments testing robustness of the Greek-adjacent result · Glossa Lab

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1. Executive Summary

This report presents seven experiments designed to test whether the Greek-dominant result in the Linear A phoneme-level analysis is an artifact of using Linear B-derived phonetic assignments. The experiments use real tablet data from John G. Younger's transcriptions (5,379 sign tokens across HT, KH, ZA, PH, KN, and other sites).

Key finding: The Greek advantage is **entirely driven by vocabulary matching**. Under scoring modes that remove vocabulary evidence (bigram+Kandles-only, Kandles-only), Greek ranks **last** of the four hypotheses. The Kandles phonetic fingerprint **marginally favours Luwian** (9.94 vs Greek 9.52). This confirms that the vocabulary-matching component is partially circular, and that the vocabulary-independent phonological signal does not support the Greek-adjacent interpretation.

What survives: Greek wins on the raw tablet corpus (margin 39.9) with full scoring. The margin is large and consistent across all large sites (HT, KH, ZA, PH). But this advantage is attributable to vocabulary matching, not to independent phonological structure.

2. Experiment 1 — Raw Tablet Sequence Replication

Hypothesis engine run on actual tablet-order sequences from corpus_manifest.csv, partitioned by archaeological site. Full scoring (bigram + Kandles + vocabulary).

Site	N tokens	Greek	Hurrian	Luwian	Semitic	Winner	Margin
ALL	5379	56.9	17.0	17.0	17.0	greek	39.9
HT	3328	56.9	16.9	16.8	16.9	greek	40.0
KH	480	23.5	15.7	15.6	15.7	greek	7.8
ZA	673	25.9	16.9	16.9	16.8	greek	8.9
PH	272	20.4	11.9	11.7	11.8	greek	8.5
KN	158	18.5	9.7	9.7	9.8	greek	8.7
ARKH	157	8.6	9.7	9.7	9.8	semitic	0.1
MA	67	7.5	8.0	8.1	7.9	luwian	0.1
TY	77	7.6	8.2	8.0	8.1	hurrian	0.1

Table A. Raw-corpus hypothesis ranking by site (full scoring). Large corpora (n≥200, highlighted) consistently return Greek as winner with margins 8–40. Small sites (ARKH, MA, TY, n<200) show no clear winner — small-sample noise.

3. Experiment 5 — Scoring Mode Comparison (Critical)

This is the most important experiment. Three scoring modes were tested on the full raw corpus: **(A) Full** = bigram + Kandles + vocabulary matches; **(B) No-vocab** = bigram + Kandles only; **(C) Kandles only**. The vocabulary-independent modes reveal whether Greek signal survives without the potentially circular lexical component.

Mode	Greek	Hurrian	Luwian	Semitic	Winner	Interpretation
A: Full (bigram+Kandles+vocab)	56.90	16.97	16.99	16.98	greek	Greek dominant — includes vocabu
B: No vocab (bigram+Kandles)	16.90	16.97	16.99	16.98	luwian	Luwian ranks #1 — Greek LAST wi
C: Kandles only	9.52	9.83	9.94	9.92	luwian	Luwian ranks #1 by Kandles fingerp

Table D. Scoring mode comparison. Green = Greek wins; Amber = Greek loses. The vocabulary-independent modes (B and C) both rank Greek last. The entire Greek advantage is attributable to vocabulary matching.

Finding: Removing vocabulary evidence (Mode B) drops Greek from score 56.9 to 16.9 — a reduction of nearly 40 points. Luwian, Semitic, and Hurrian are essentially unaffected (they have no vocabulary bonus). Under Kandles-only scoring (Mode C), Greek phonetic fingerprint (9.52) ranks below Luwian (9.94) and Semitic (9.92). This directly confirms that the vocabulary component is circular: the vocabulary list was derived by applying Linear B phonetic values, and the engine recovers those same assignments.

4. Experiment 2 — Mapping Ablation

The phoneme mapping was progressively reduced to only the top-10/20/30/40 and all available signs (104). At each level, 30 trials with random sign subsets were run under no-vocab scoring. This tests whether the Greek advantage (if any) grows as more mapping is applied.

N signs	Greek mean	95% CI	Hurrian mean	Greek #1 fraction
10	16.878	[16.868, 16.890]	16.942	13%
20	16.896	[16.881, 16.910]	16.899	30%
30	16.899	[16.881, 16.918]	16.818	57%
40	16.891	[16.871, 16.910]	16.760	83%
104	16.903	[16.903, 16.903]	16.966	0%

Table B. Mapping ablation results (no-vocab scoring). Greek #1 fraction = proportion of 30 trials where Greek ranked first. At 40 signs: Greek #1 in 83% of trials (consistent with a marginal advantage).

Finding: Greek rank-1 fraction increases from 13% (n=10 signs) to 83% (n=40 signs), suggesting that the full mapping does confer a slight bigram advantage independent of vocabulary. However, the mean scores are all within ~0.1 of each other — the effect is tiny and not robust to the no-vocab test.

5. Experiment 4 — Random Mapping Null Distribution

The real mapping score was compared against distributions from (a) frequency-matched random mappings, (b) CV-structure-preserving random, (c) permuted Linear B correspondences. 30 trials each, no-vocab scoring.

Control type	Trials	Null mean	Real score	p-value	z-score	Interpretation
frequency matched random	30	16.889	16.903	0.4000	0.292	Real ≈ null; mapping structure unimportant
cv structure preserving	30	16.891	16.903	0.4000	0.269	Real ≈ null; CV structure unimportant
permuted lb correspondences	30	16.889	16.903	0.4000	0.292	Real ≈ null; specific LB mapping unimportar

Table C. Random/permuted mapping null distribution (no-vocab scoring). Real Greek score = 16.903. All p-values ≈ 0.40, z-scores ≈ 0.29. Real mapping is NOT statistically distinguishable from random.

Finding: Under no-vocab scoring, the real Linear B phoneme mapping produces a Greek score that is statistically indistinguishable from random mappings (p≈0.40, z≈0.29). This means the specific sign-to-phoneme correspondences do not confer a detectable advantage in bigram+Kandles scoring. The only statistically significant Greek advantage comes from vocabulary matching.

6. Experiment 7 — Null Corpus Controls

Corpus type	Greek mean	95% CI	Notes
real	16.903	[16.903, 16.903]	Actual tablet sequence — baseline
shuffled	16.931	[16.919, 16.941]	Sign order randomised — Greek HIGHER (not lower!)
unigram_only	16.924	[16.915, 16.935]	Bigrams destroyed — Greek HIGHER

Table E. Null corpus controls (no-vocab scoring). Shuffled and unigram corpora produce marginally higher Greek scores than the real corpus.

Finding: Greek score does not decrease when corpus structure is destroyed — it actually increases slightly for shuffled and unigram corpora. This confirms that the no-vocab Greek score (~16.9) is driven by random baseline effects (all four hypotheses score ~16.9–17.0 regardless of corpus structure), not by meaningful sequential patterns in the real tablet data.

7. Experiment 3 — Mapping Perturbation (No-Vocab)

Noise level	Greek mean	95% CI
0%	16.903	[16.903, 16.903]
5%	16.901	[16.892, 16.909]
10%	16.899	[16.888, 16.910]
15%	16.892	[16.873, 16.908]
20%	16.885	[16.869, 16.900]
30%	16.906	[16.891, 16.921]

Table F. Perturbation results. Greek score is stable across all noise levels because the no-vocab signal is essentially zero — noise has nothing to destroy.

8. Experiment 6 — Language Model Fairness

Equalised all language model corpora to 2000 characters (baseline Greek model was substantially larger). Winner: baseline=luwian, equalized=luwian. Greek rank: baseline=#4, equalized=#4. Greek still ranks last under both conditions (no-vocab scoring), confirming that model size is not the driver of the non-Greek result.

9. Integrated Interpretation

The anti-circularity experiments produce a coherent and scientifically honest picture:

• **Greek wins with vocabulary (Exp 1, 5A):** On real tablet data with full scoring, Greek wins all large-corpus splits by margins of 8–40 points. This is real and consistent.

• **Vocabulary matching is the mechanism (Exp 5B/C):** Remove vocabulary, and Greek drops to last place in both bigram and Kandles scoring. The vocabulary component accounts for ~40 of the ~40-point margin. This directly confirms circularity.

• **Kandles favours Luwian (Exp 5C):** The vocabulary-independent phonetic fingerprint marginally favours Luwian (9.94) over Greek (9.52). This is a novel finding that partially supports the Anatolian hypothesis.

• **Mapping structure irrelevant to bigram+Kandles (Exp 4):** The real Linear B correspondence mapping is statistically indistinguishable from random or permuted mappings under no-vocab scoring ($p \approx 0.40$). The specific sign assignments do not drive the phonological signal.

• **Null corpus confirms baseline-level scoring (Exp 7):** Shuffled and unigram corpora produce equivalent or higher Greek scores — confirming the ~16.9 baseline is noise-level, not signal.

• **Mapping ablation shows weak ordering (Exp 2):** Greek rank-1 fraction increases from 13% to 83% as more signs are included, suggesting a very weak genuine bigram signal — but not strong enough to survive scoring without vocabulary.

9.1 Conservative conclusion

The Greek-adjacent result in the full-scoring analysis is **primarily driven by vocabulary matching, and that vocabulary matching is partially circular**. The vocabulary list (ku-ro, ki-re-ta, sa-ra2, etc.) was derived by applying Linear B phonetic values to Linear A signs; the engine then finds those same values in the corpus and counts it as evidence for Greek.

The vocabulary-independent phonological signal (Kandles fingerprint) does not favour Greek — it marginally favours Luwian/Anatolian. This does not prove Minoan is Anatolian, but it does mean the Greek-adjacent phonological claim cannot be supported independently of the circular vocabulary evidence.

9.2 What would be needed for a stronger claim

• An independent vocabulary source not derived from Linear B phonetic values (e.g., confirmed loanwords, bilingual texts, or parallel inscriptions).

• Fuller and validated language models for Hurrian and Luwian at phoneme level (the current models are minimal character-level corpora).

• Site-by-site analysis showing consistent Kandles ranking across HT, KH, and ZA.

• A proper null model for the Kandles phonetic fingerprint to assess statistical significance of the Luwian>Greek ordering.

• The ICIT corpus (Fuls/Wells) for validation on the full Mahadevan sign inventory.

References

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